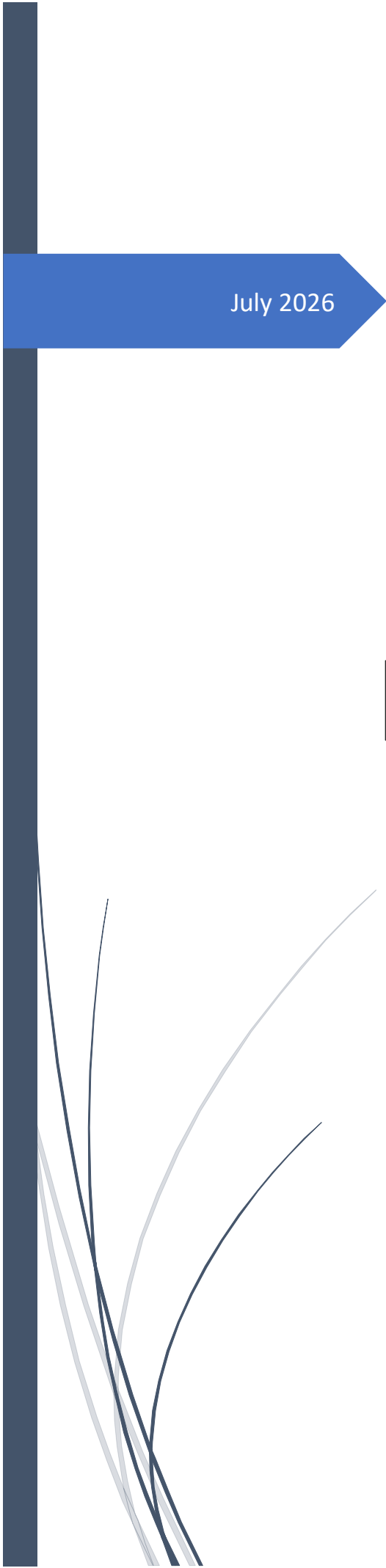




July 2026

Bank Customer Churn Analysis

Customer Behaviour, Churn Drivers & Retention Insights

Dataset Source: Kaggle – Thomas Konstantin
Tools Used: Python | SQL | Power BI

Nikhil Kumar

Data Analytics Portfolio Project

Table of Contents

1. Executive Summary	4
2. Business Problem & Objectives	
2.1 Business Problem	4
2.2 Project Objectives	4
2.3 Key Business Question	4, 5
3. Dataset Overview	
3.1 Data Source	5
3.2 Dataset Size	5
3.3 Target Variable	5
3.4 Key Variables	5, 6
4. Data Preparation & Cleaning	
4.1 Initial Data Inspection	6
4.2 Missing Values and Duplicates	6, 7
4.3 Data Quality and Consistency Checks	7
4.4 Preparation for Further Analysis	7
5. Exploratory Data Analysis & Key Findings	
5.1 Overall Customer Churn	7
5.2 Customer Demographics	7, 8
5.3 Transaction Activity	8
5.4 Customer Relationships	8
5.5 Customer Inactivity	8
5.6 Customer Contact Frequency	8, 9
5.7 Combined Behavioural Risk Indicators	9
5.8 Existing Customers Requiring Further Attention	10
6. SQL Business Analysis	
6.1 Business Questions Analysed	10, 11
6.2 Behavioural Risk Segmentation	11
6.3 Contribution to Total Churn	11
6.4 Identification of Existing Higher-Risk Customers	11, 12
6.5 Role of SQL in the Project	12
7. Power BI dashboard	
7.1 Key Performance Indicators	12
7.2 Dashboard Visualisations	12
7.3 Interactive Customer Segmentation	13

7.4 Final Dashboard.....	13
7.5 Dashboard Purpose	13
8. Key Business Insights	
8.1 Transaction Activity is a Strong Churn Indicator	14
8.2 Customer with Fewer Banking Relationships Show Higher Churn	14
8.3 Inactivity and Frequent Contact May Signal Declining Engagement	14
8.4 Multiple Behavioural Indicators Provide a Clearer Risk Signal	14
8.5 High Churn Rate and High Churn Volume are Not the Same.....	14, 15
8.6 Existing Customers with Multiple Warning Indicators Provide an Actionable Segment	15
9. Business Recommendations	
9.1 Develop a Behavioural Early-Warning Framework.....	15
9.2 Prioritise Existing Customers with Multiple Risk Indicators	15
9.3 Monitor Declining Transaction Activity	15, 16
9.4 Strengthen Customer Relationships.....	16
9.5 Investigate Repeated Customer Contacts.....	16
9.6 Introduce Inactivity-Based Engagement Triggers.....	16
9.7 Measure Retention Strategy Effectiveness	16
10. Limitations & Future Analysis	
10.1 Descriptive Rather than Predictive Analysis	17
10.2 Association Does Not Establish Causation.....	17
10.3 Limited Customer Experience Information.....	17
10.4 Limited Behavioural History	17
10.5 Behavioural Risk Thresholds.....	18
10.6 Future Predictive Analysis	18
10.7 Retention Strategy Evaluation	18
11. Conclusion	18,19
12. Reference	19

1. Executive Summary

This project analyses bank customer data to identify the key behavioural and demographic factors associated with customer churn. The analysis focuses on understanding how transaction activity, customer inactivity, relationship depth, contact frequency, income category, card category, and other risk indicators relate to customer attrition. The dataset contains 10,127 customers, of whom 1,627 attrited and 8,500 remained, resulting in an overall churn rate of 16.07%.

The analysis was conducted using data analytics techniques and presented through an interactive Power BI dashboard. The dashboard enables users to explore churn patterns across different customer segments using Income Category, Card Category, and Gender filters. The findings highlight customer behaviours and risk indicators associated with higher churn, providing insights that can support more targeted customer retention strategies.

2. Business Problem & Objectives

2.1 Business Problem

Customer churn represents a significant challenge for banks because losing existing customers can reduce revenue, weaken customer relationships, and increase the cost of acquiring replacement customers. Understanding which customers are more likely to leave—and the behaviours associated with attrition—can help banks develop more targeted retention strategies. This project aims to analyse customer characteristics and behavioural patterns to identify the major factors associated with churn and highlight customer segments that may require greater retention attention.

2.2 Project Objectives

1. Measure the overall customer churn rate and compare attrited and existing customers.
2. Analyse how transaction activity relates to customer churn.
3. Examine whether customer inactivity is associated with higher churn.
4. Investigate the relationship between customer relationships and churn.
5. Assess how customer contact frequency relates to churn behaviour.
6. Compare churn rates across different income categories.
7. Identify how combinations of churn-related factors, represented by the Risk Indicator Count, relate to customer attrition.
8. Develop an interactive Power BI dashboard that allows users to explore churn patterns using Income Category, Card Category, and Gender filters.
9. Generate actionable insights that could help a bank identify higher-risk customer groups and support customer retention strategies.

2.3 Key Business Question

The overall question guiding the project is:

What factors are associated with bank customer churn, and which customer segments should be prioritised for retention efforts?

3. Dataset Overview

3.1 Data Source

The dataset used in this project was obtained from Kaggle and contains information about credit card customers, including customer demographics, account relationships, transaction behaviour, credit usage, and customer attrition status. The dataset contains customer-level variables such as age, gender, education level, income category, card category, credit limit, transaction activity, inactivity, and credit utilisation. The `Attrition_Flag` variable identifies whether a customer remained with the bank or became an attrited customer.

Data Source: Kaggle – Thomas Konstantin

Dataset File: BankChurners.csv

Kaggle Reference: [Bank Churn Data Exploration And Churn Prediction](#)

3.2 Dataset Size

The dataset contains:

1. 10,127 customer records
2. 21 variables

Each row represents an individual customer, while the columns contain information relating to customer demographics, banking relationships, account activity, transaction behaviour, and financial characteristics. The cleaned dataset contained no missing values or duplicate records.

3.3 Target Variable

The primary variable used to analyse customer churn is:

Attrition_Flag

Customers are classified into two groups:

1. Existing Customer — customer remained with the bank.
2. Attrited Customer — customer churned.

Of the 10,127 customers, 1,627 are attrited customers and 8,500 are existing customers, producing an overall churn rate of 16.07%.

3.4 Key Variables

The analysis focused on several groups of variables:

1. Customer demographics: Customer Age, Gender, Education Level, Marital Status and Income Category.
2. Customer relationship and engagement: Months on Book, Total Relationship Count, Months Inactive in the Last 12 Months and Contacts Count in the Last 12 Months.
3. Financial characteristics: Credit Limit, Total Revolving Balance, Average Open to Buy and Average Utilisation Ratio.
4. Transaction behaviour: Total Transaction Amount, Total Transaction Count, Transaction Amount Change and Transaction Count Change.
5. Account characteristics: Card Category and Dependent Count.

These variables were analysed to identify differences between existing and attrited customers and determine which characteristics were most strongly associated with customer churn.

4. Data Preparation & Cleaning

Before conducting the churn analysis, the dataset was inspected and prepared in Python using Google Colab to ensure that the data was suitable for analysis.

4.1 Initial Data Inspection

An initial assessment was performed to understand the structure and quality of the dataset. This included reviewing:

1. Dataset dimensions and column names
2. Data types
3. Summary statistics
4. Missing values
5. Duplicate records
6. Unique values within categorical variables
7. Numerical distributions and potential outliers

This initial inspection helped identify potential data-quality issues before beginning the exploratory and business analysis.

4.2 Missing Values and Duplicates

The dataset was checked for missing values and duplicate records to ensure that incomplete or repeated observations would not affect the analysis.

Following the data-preparation process, the final cleaned dataset contained:

10,127 customer records

21 variables

No missing values

No duplicate records

This cleaned dataset was subsequently used for the SQL analysis and Power BI dashboard.

4.3 Data Quality and Consistency Checks

Categorical and numerical variables were reviewed for inconsistencies and unusual values. Key customer attributes such as gender, education level, income category, card category, customer status, transaction activity, credit information, and account engagement were examined before further analysis. These checks helped ensure that the values used in subsequent calculations and visualisations were consistent and analytically usable.

4.4 Preparation for Further Analysis

After completing the cleaning and validation process, the prepared dataset was exported for the next stages of the project.

The analytical workflow followed:

Raw Dataset → Python Data Cleaning & EDA → Cleaned Dataset → SQL Analysis → Power BI Dashboard → Business Insights

This approach allowed each tool to serve a specific purpose: Python for preparing and exploring the data, SQL for detailed business analysis, and Power BI for interactive visualisation and communication of the results.

5. Exploratory Data Analysis & Key Findings

Exploratory data analysis was conducted to understand customer characteristics, compare existing and attrited customers, and identify patterns associated with customer churn. The analysis considered demographic, financial, transactional, and customer-engagement characteristics.

5.1 Overall Customer Churn

The dataset contains 10,127 customers, including 1,627 attrited customers and 8,500 existing customers.

This represents an overall customer churn rate of:

16.07%

This means approximately one in six customers in the dataset had churned, providing the baseline for comparing churn across different customer groups.

5.2 Customer Demographics

Demographic characteristics such as age, gender, education level, marital status, and income category were examined to determine whether particular customer groups experienced noticeably higher churn. Some variation was observed between customer segments; however, demographic characteristics generally provided weaker churn differentiation than customer behavioural variables. For example, the average age of attrited customers was approximately

46.66 years, compared with 46.26 years among existing customers, indicating very little difference in average age between the two groups.

This suggests that demographic characteristics alone may have limited value for identifying customers displaying higher churn risk.

5.3 Transaction Activity

Transaction behaviour emerged as one of the clearest differentiators between existing and attrited customers. Attrited customers completed an average of approximately 44.93 transactions, compared with 68.67 transactions among existing customers.

Average transaction amount also differed considerably:

Attrited Customers: 3,095.03

Existing Customers: 4,654.66

Customers completing 31–50 transactions recorded a churn rate of approximately 40.45%, while customers with higher transaction activity generally experienced substantially lower churn. These results indicate that lower transaction activity is strongly associated with customer churn.

5.4 Customer Relationships

The number of relationships customers maintained with the bank also showed an important churn pattern.

Customers with:

1 relationship - 25.60% churn

2 relationships - 27.84% churn

6 relationships - 10.50% churn

Overall, customers with fewer banking relationships tended to experience higher churn. This suggests that customers with broader relationships across the bank's products and services may demonstrate stronger retention.

5.5 Customer Inactivity

Customer inactivity was another important engagement characteristic. Churn generally increased across several levels of inactivity, suggesting that customers who become less engaged with their accounts may represent a useful group for retention monitoring, however, the relationship was not perfectly linear across every inactivity category. Therefore, inactivity should be considered alongside other behavioural characteristics rather than used as an isolated churn indicator.

5.6 Customer Contact Frequency

Higher customer contact frequency was associated with increased churn.

For example:

0 contacts → 1.75% churn
3 contacts → 20.15% churn
5 contacts → 33.52% churn
6 contacts → 100% churn

The six-contact group contained only 54 customers, so the 100% churn rate should be interpreted cautiously because of the small group size. The overall pattern suggests that repeated customer contact may be associated with dissatisfaction, unresolved service issues, or other circumstances related to churn. However, the dataset does not contain the reasons for customer contact, so the relationship should not be interpreted as causal.

5.7 Combined Behavioural Risk Indicators

Four behavioural characteristics were combined to create a descriptive Risk Indicator Count:

1. Total transaction count of 50 or fewer
2. Total relationship count of 2 or fewer
3. Contact count of 3 or more
4. Inactive months of 3 or more

A strong increase in churn was observed as customers accumulated these indicators:

Risk Indicators	Churn Rate
0	1.57%
1	6.24%
2	18.33%
3	48.36%
4	96.63%

This was one of the strongest findings in the project. Customers displaying multiple behavioural warning indicators had substantially higher observed churn rates than customers displaying few or none. The indicators are descriptive analytical rules rather than a predictive machine-learning model, so they should be used to identify segments for further investigation rather than interpreted as individual churn probabilities.

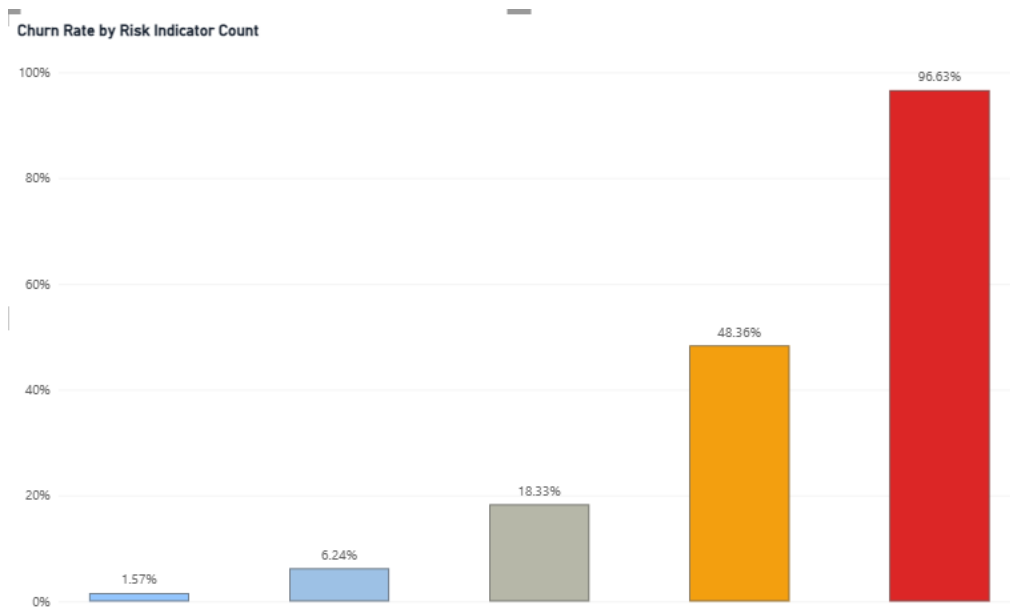


Figure 1. Churn Rate by Number of Behavioural Risk Indicators

5.8 Existing Customers Requiring Further Attention

Applying the behavioural indicators to customers who had not yet churned identified 619 existing customers with at least three risk indicators. Among these customers, 6 displayed all four indicators. These customers could represent useful priority segments for further retention analysis because they currently remain with the bank while displaying several characteristics associated with historically higher churn.

6. SQL Business Analysis

Following data preparation and exploratory analysis, the cleaned dataset was imported into DB Browser for SQLite for more detailed business analysis. SQL queries were developed to investigate customer churn across demographic, financial, transactional, and behavioural characteristics. The analysis also used conditional logic, aggregations, grouping, subqueries, and Common Table Expressions (CTEs) to develop customer risk segments and identify existing customers displaying multiple behavioural warning indicators.

6.1 Business Questions Analysed

The SQL analysis addressed a range of business questions, including:

1. What is the overall customer churn rate?
2. How does churn vary across demographic and income groups?
3. How does transaction activity differ between existing and attrited customers?
4. Does churn increase as customer inactivity increases?
5. How does the number of customer relationships relate to churn?
6. Is customer contact frequency associated with higher churn?

7. How do financial characteristics such as credit limit and available credit differ between existing and attrited customers?
8. How does churn vary across card categories?
9. Does customer tenure or age meaningfully differentiate attrited customers?
10. How does churn change when customers accumulate multiple behavioural risk indicators?
11. Which existing customers currently display multiple behavioural warning indicators?

6.2 Behavioural Risk Segmentation

One of the key SQL analyses combined four behavioural characteristics into a Risk Indicator Count.

Customers received one indicator for each of the following conditions:

1. Total_Trans_Ct <= 50
2. Total_Relationship_Count <= 2
3. Contacts_Count_12_mon >= 3
4. Months_Inactive_12_mon >= 3

The resulting segmentation demonstrated a substantial increase in churn as the number of indicators increased. Customers with no indicators had a churn rate of only 1.57%, compared with 18.33% for customers with two indicators, 48.36% for customers with three indicators, and 96.63% for customers displaying all four indicators.

6.3 Contribution to Total Churn

The analysis also examined how much each risk group contributed to the overall population of attrited customers. Customers displaying two risk indicators accounted for 37.80% of total churn, while customers displaying three indicators accounted for 35.28%. Together, these two groups represented 73.08% of all attrited customers. Although customers displaying all four indicators experienced the highest churn rate, they represented only 10.57% of total churn because this segment contained substantially fewer customers. This distinction is important because the segment with the highest churn rate is not necessarily the segment responsible for the largest number of churned customers.

6.4 Identification of Existing Higher-Risk Customers

The behavioural framework was subsequently applied to customers who were still classified as existing customers. The SQL analysis identified 619 existing customers with at least three behavioural risk indicators, including 6 customers displaying all four indicators. This provides a potentially actionable customer segment for further investigation. Rather than applying retention initiatives equally across the entire customer base, the bank could use behavioural segmentation to prioritise customers displaying several warning signals. However, these

indicators are based on descriptive historical relationships and should not be interpreted as predictions that individual customers will churn.

6.5 Role of SQL in the Project

SQL provided a structured method for translating exploratory observations into measurable business insights. It enabled customer groups to be compared systematically, churn rates to be calculated across different characteristics, and behavioural conditions to be combined into more meaningful customer segments. The SQL analysis therefore acted as the bridge between the initial exploratory analysis and the final Power BI dashboard and business recommendations.

7. Power BI Dashboard

An interactive Power BI dashboard was developed to communicate the key findings of the customer churn analysis in a clear and accessible format. The dashboard brings together the major KPIs, behavioural churn patterns, customer segmentation, and interactive filtering capabilities identified throughout the project.

7.1 Key Performance Indicators

Four KPI cards provide an immediate overview of customer attrition:

1. Total Customers – 10,127
2. Attrited Customers – 1,627
3. Existing Customer – 8,500
4. Churn Rate – 16.07%

These measures provide a high-level summary of the customer base and establish the overall churn position before users explore individual customer segments.

7.2 Dashboard Visualisations

The dashboard contains six analytical visualisations:

1. Churn Rate by Transaction Activity
2. Churn Rate by Customer Inactivity
3. Churn Rate by Income Category
4. Churn Rate by Customer Relationships
5. Churn Rate by Customer Contact Frequency
6. Churn Rate by Risk Indicator Count

The visualisations focus primarily on customer behaviour and engagement because these characteristics produced some of the clearest churn patterns during the analysis. The Risk Indicator Count visual uses progressive colour formatting to emphasise the increase in churn as customers accumulate multiple behavioural warning indicators.

7.3 Interactive Customer Segmentation

Three interactive slicers were incorporated into the dashboard:

1. Income Category
2. Card Category
3. Gender

These filters allow users to dynamically explore how customer counts, churn rates, and behavioural patterns change across different customer segments. For example, selecting an individual card or income category automatically updates the KPI cards and analytical visualisations, allowing users to investigate churn within specific customer groups.

7.4 Final Dashboard



Figure 2. Bank Customer Churn Analysis — Interactive Power BI Dashboard

7.5 Dashboard Purpose

The dashboard provides a concise decision-support view of customer churn by combining overall performance indicators with customer-level behavioural patterns. Rather than presenting every variable in the dataset, the dashboard focuses on characteristics that provide useful analytical insights into customer attrition. The interactive filtering functionality also allows users to move beyond overall averages and investigate churn within specific customer segments. This makes the dashboard suitable for supporting exploratory retention analysis and communicating the key findings to both technical and non-technical stakeholders.

8. Key Business Insights

The analysis identified several important patterns associated with customer attrition. Overall, behavioural and engagement characteristics provided stronger churn signals than many individual demographic or financial characteristics.

8.1 Transaction Activity Is a Strong Churn Indicator

Lower transaction activity was strongly associated with customer churn. Attrited customers completed an average of 44.93 transactions, compared with 68.67 transactions among existing customers. Customers completing 31–50 transactions experienced a churn rate of 40.45%, indicating that reduced transaction activity may provide an important signal of declining customer engagement.

8.2 Customers with Fewer Banking Relationships Show Higher Churn

Customers maintaining fewer relationships with the bank generally experienced higher churn. Customers with one relationship had a churn rate of 25.60%, while customers with two relationships had a churn rate of 27.84%. In comparison, customers with six relationships recorded a churn rate of 10.50%. This suggests that stronger and broader customer relationships may be associated with greater customer retention.

8.3 Inactivity and Frequent Contact May Signal Declining Engagement

Customer inactivity and contact frequency both demonstrated meaningful relationships with churn. In particular, churn generally increased with higher contact frequency, reaching 33.52% among customers with five contacts. This could indicate unresolved service issues or dissatisfaction, although the dataset does not provide the reason for each customer contact. These behaviours may therefore provide useful signals for identifying customers requiring further attention.

8.4 Multiple Behavioural Indicators Provide a Clearer Risk Signal

The strongest pattern emerged when multiple behavioural indicators were considered together.

Churn increased from:

1.57% → 6.24% → 18.33% → 48.36% → 96.63%

As the number of behavioural risk indicators increased from zero to four. This suggests that combining multiple behavioural characteristics can provide a more useful view of churn concentration than relying on individual indicators alone.

8.5 High Churn Rate and High Churn Volume Are Not the Same

Customers displaying all four behavioural indicators experienced the highest churn rate at 96.63%. However, this relatively small segment accounted for only 10.57% of total churn. In comparison, customers displaying two and three indicators accounted for 73.08% of all

attrited customers combined. This distinction is important for retention planning because the segment with the highest individual churn rate may not necessarily represent the greatest overall retention opportunity.

8.6 Existing Customers with Multiple Warning Indicators Provide an Actionable Segment

The analysis identified 619 existing customers displaying at least three behavioural risk indicators, including 6 customers displaying all four indicators. These customers provide a potential priority segment for further investigation and targeted retention initiatives. However, the behavioural indicators are descriptive rather than predictive and should not be interpreted as individual churn probabilities.

9. Business Recommendations

Based on the customer churn patterns identified in the analysis, the following recommendations could help the bank prioritise retention efforts and improve customer engagement.

9.1 Develop a Behavioural Early-Warning Framework

The bank could monitor the four behavioural indicators identified in the analysis:

1. Low transaction activity
2. Fewer customer relationships
3. Higher contact frequency
4. Increased customer inactivity

Customers displaying multiple indicators could be flagged for further investigation by customer-retention teams. The analysis showed that churn increased from 1.57% among customers with no indicators to 48.36% among customers with three indicators and 96.63% among customers with all four indicators. This framework should initially be treated as a descriptive screening tool rather than a predictive churn model.

9.2 Prioritise Existing Customers with Multiple Risk Indicators

The analysis identified 619 existing customers with at least three behavioural risk indicators, including 6 customers displaying all four. Rather than applying the same retention strategy across the entire customer base, these customers could be prioritised for further investigation and appropriate retention interventions. Customers displaying three indicators may be particularly important because they represent a larger group while still being associated with a substantially elevated historical churn rate.

9.3 Monitor Declining Transaction Activity

Lower transaction activity was one of the strongest characteristics associated with churn. The bank could monitor changes in transaction frequency over time and identify customers whose

activity declines substantially relative to their previous behaviour. Depending on the customer's circumstances, appropriate interventions could include personalised communications, account reviews, relevant product benefits, or targeted engagement campaigns.

9.4 Strengthen Customer Relationships

Customers with fewer banking relationships generally experienced higher churn. Where appropriate to the customer's financial needs, the bank could improve awareness and adoption of relevant products and services to strengthen the overall customer relationship. Any cross-selling strategy should remain customer-focused and needs-based, rather than simply encouraging customers to hold additional products.

9.5 Investigate Repeated Customer Contacts

Higher contact frequency was associated with increased churn, with customers recording five contacts experiencing a churn rate of 33.52%. The bank could investigate customers making repeated contacts to determine whether they are experiencing unresolved complaints, service difficulties, account issues, or other problems. Additional data capturing the reason, channel, outcome, and resolution status of customer contacts would help the bank better understand this relationship.

9.6 Introduce Inactivity-Based Engagement Triggers

Customer inactivity could be incorporated into ongoing engagement monitoring. Customers reaching defined inactivity thresholds could trigger an account review or appropriate communication before disengagement becomes more significant. Rather than assuming that inactivity automatically indicates churn, these triggers could be combined with transaction activity, relationship count, and contact frequency to provide a more complete view of customer engagement.

9.7 Measure Retention Strategy Effectiveness

Retention initiatives should be evaluated to determine whether they actually improve customer outcomes.

The bank could monitor measures such as:

1. Retention rate after intervention
2. Transaction activity
3. Customer re-engagement
4. Contact resolution
5. Campaign response

Controlled testing or comparable customer groups could also be used to determine which interventions provide the greatest improvement in retention.

10. Limitations & Future Analysis

Although the analysis identified several meaningful patterns associated with customer churn, there are limitations that should be considered when interpreting the findings.

10.1 Descriptive Rather Than Predictive Analysis

The behavioural Risk Indicator Count developed in this project is based on patterns observed within the existing dataset. It is therefore a descriptive analytical framework rather than a predictive churn model. Customers displaying multiple risk indicators historically experienced substantially higher churn rates, but this does not mean that every existing customer displaying these characteristics will eventually churn. A future extension of the project could develop and evaluate a machine-learning classification model to estimate individual customer churn probabilities.

10.2 Association Does Not Establish Causation

The analysis identifies relationships between customer characteristics and churn but does not establish that these characteristics directly cause customers to leave the bank. For example, higher customer contact frequency was associated with higher churn. However, the dataset does not explain the reasons for these contacts. Customers may have contacted the bank because of complaints, account issues, service requests, or other circumstances. Additional information would therefore be required before determining whether particular behaviours directly contribute to customer attrition.

10.3 Limited Customer Experience Information

The dataset primarily contains demographic, financial, account, and behavioural information. It does not provide detailed customer-experience information such as:

1. Customer satisfaction scores
2. Complaints and service issues
3. Reasons for contacting the bank
4. Contact resolution outcomes
5. Competitor offers
6. Reasons for account closure

Including these variables could provide a more complete understanding of the factors contributing to customer churn.

10.4 Limited Behavioural History

Several behavioural variables represent aggregated customer activity rather than a detailed chronological history. For example, total transaction activity provides useful information about customer engagement but does not show exactly when activity started declining. Future analysis using monthly or time-series customer behaviour could help identify changes in transaction activity, inactivity, or engagement before a customer churns.

10.5 Behavioural Risk Thresholds

The thresholds used to construct the Risk Indicator Count were developed from patterns observed during the analysis, including low transaction activity, fewer relationships, higher contact frequency, and greater inactivity. These thresholds have not been independently validated on another customer dataset. Their effectiveness should therefore be tested on additional data before being used operationally for customer-retention decisions.

10.6 Future Predictive Analysis

The project could be extended by developing a customer churn prediction model using techniques such as:

1. Logistic Regression
2. Decision Tree
3. Random Forest

Model performance could be evaluated using appropriate classification metrics such as precision, recall, F1-score, ROC-AUC, and a confusion matrix. Feature importance or model interpretation techniques could then be used to determine which customer characteristics contribute most strongly to churn predictions.

10.7 Retention Strategy Evaluation

A further extension would be to evaluate whether retention interventions actually reduce customer churn. Customers identified for retention attention could be divided into appropriate comparison groups, allowing the bank to measure whether specific interventions improve customer retention, engagement, or transaction activity. This would move the analysis beyond identifying churn-related patterns toward determining which business actions actually improve customer outcomes.

11. Conclusion

This project analysed bank customer churn to identify the demographic, financial, and behavioural characteristics associated with customer attrition. Using an end-to-end analytics workflow, customer data was prepared and explored before being analysed in greater depth and presented through an interactive dashboard. The dataset contained 10,127 customers, including 1,627 attrited customers, resulting in an overall churn rate of 16.07%.

The analysis found that customer behaviour and engagement provided some of the clearest churn signals. Lower transaction activity, fewer banking relationships, greater inactivity, and higher customer contact frequency were associated with increased churn. In comparison, characteristics such as customer age and tenure showed relatively small differences between existing and attrited customers.

Combining four behavioural characteristics into a Risk Indicator Count provided an additional perspective on customer attrition. Churn increased from 1.57% among customers with no behavioural risk indicators to 96.63% among customers displaying all four indicators. The

analysis also identified 619 existing customers displaying at least three indicators, highlighting a potential customer segment for further investigation and targeted retention initiatives. These indicators remain descriptive and would require further validation before being used as a predictive decision-making tool.

Overall, the project demonstrates how Python, SQL, and Power BI can be combined to transform raw customer data into meaningful business insights. Python supported data preparation and exploratory analysis, SQL enabled detailed customer segmentation and business-focused investigation, and Power BI provided an interactive method for communicating the findings. The resulting insights could support more targeted customer-retention strategies, while future predictive modelling and additional customer-experience data could further strengthen the bank's ability to identify and understand customers at risk of churn.

12. Reference

Konstantin, T. (n.d.). *Bank Churn Data Exploration and Churn Prediction*. Kaggle. (<https://www.kaggle.com/code/thomaskonstantin/bank-churn-data-exploration-and-churn-prediction/input>).